Unit: 6 Unit 7: Factoring
Topic: Expanding Special Patterns Perfect Square Trinomial

Name: ________________
Date: ____________

Challenge (Habanero)

Q1. What is the expansion of $(x+2)^2$?
(A) x^2+4x+4
(B) x^2+2x+4
(C) x^2-4x+4
(D) x^2+4x-4
(E) x^2-2x-4

Q2. A hiker is traveling at a rate of x miles per hour. If the hiker's speed is increased by 3 miles per hour, what is the expansion of $(x+3)^2$?
(A) x^2+6x+9
(B) x^2+3x+9
(C) x^2-6x+9
(D) x^2+6x-9
(E) x^2-3x-9

Q3. What is the factored form of the perfect square trinomial $x^2+10x+25$?
(A) $(x+5)^2$
(B) $(x-5)^2$
(C) $(x+2)^2$
(D) $(x-2)^2$
(E) $(x+3)^2$

Q4. A park ranger is tracking the population of a species of animal. If the population is currently x and is increasing at a rate of 4 animals per year, what is the expansion of $(x+4)^2$?
(A) $x^2+8x+16$
(B) $x^2+4x+16$
(C) $x^2-8x+16$
(D) $x^2+8x-16$
(E) $x^2-4x-16$

Q5. What is the expansion of $(2x-3)^2$?
(A) $4x^2-12x+9$
(B) $4x^2+12x+9$
(C) $4x^2-12x-9$
(D) $4x^2+12x-9$
(E) $4x^2+6x+9$

Q6. A group of friends are planning a road trip. If their car gets x miles per gallon and they drive for 2 hours at a speed of x miles per hour, what is the expansion of $(xcdot2)^2$?
(A) $4x^2$
(B) $2x^2$
(C) x^2
(D) $4x$
(E) $2x$

Q7. What is the factored form of the perfect square trinomial $4x^2+12x+9$?
(A) $(2x+3)^2$
(B) $(2x-3)^2$
(C) $(x+3)^2$
(D) $(x-3)^2$
(E) $(x+2)^2$

Q8. A farmer is harvesting crops at a rate of x bushels per hour. If the farmer's harvesting speed is increased by 2 bushels per hour, what is the expansion of $(x+2)^2$?
(A) x^2+4x+4
(B) x^2+2x+4
(C) x^2-4x+4
(D) x^2+4x-4
(E) x^2-2x-4

Open Ended Questions (FRQ)

Q9. A park is planning to build a new hiking trail. The trail will be x miles long and will have a total elevation gain of x^2+6x+9 feet. Factor the expression x^2+6x+9 and explain its significance in the context of the hiking trail.

Q10. A group of friends are planning a road trip to a national park. Their car gets x miles per gallon, and they expect to drive for $x+4$ hours. What is the expansion of $(x+4)^2$? Use this expansion to calculate the total distance they will travel, assuming they drive at a constant speed of x miles per hour.

Chef's Tips

- Tip 1: Encourage students to use the formula $(a+b)^2=a^2+2ab+b^2$ to expand special patterns.
- Tip 2: Remind students to look for perfect square trinomials in the form $a^2+2ab+b^2$ and factor them as $(a+b)^2$.
- Tip 3: Emphasize the importance of checking their work by plugging their factored expressions back into the original equation.